ACTION CHUNK SEAM BOUNDARY: DERIVATIVE CONTINUITY VS. TRANSMISSION SHOCK

Systems Dynamics of Unchecked C^0 Concatenation vs. Jerk-Continuous C^2 Quintic Spline Trajectory Blending

ACTIVE ACTION CHUNK A

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Horizon: t ∈ [0, 50 ms] (20 Hz)
q_A(50) = 0.850 rad
q̇_A(50) = −0.45 rad/s (Decel)
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VLA POLICY INFERENCE (BRAIN)

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Request Lead: T_req = 0 ms
Tail Latency: L ~ P99 = 48 ms
Chunk B arrives at t = 48 ms
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CASE A: Naive C^0 Concatenation

Pos matched; $\Delta\dot{q}$ = +0.60 rad/s step!

CASE B: C^2 Jerk-Continuous Spline

Quintic bridge over T_blend = 20 ms

1 kHz SERVO LOOP & PLANT

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Servo Period: T_c = 1.0 ms
Reflected Inertia: J = 2.7 kg·m²
Pin Shear Limit: τ = 285 N·m
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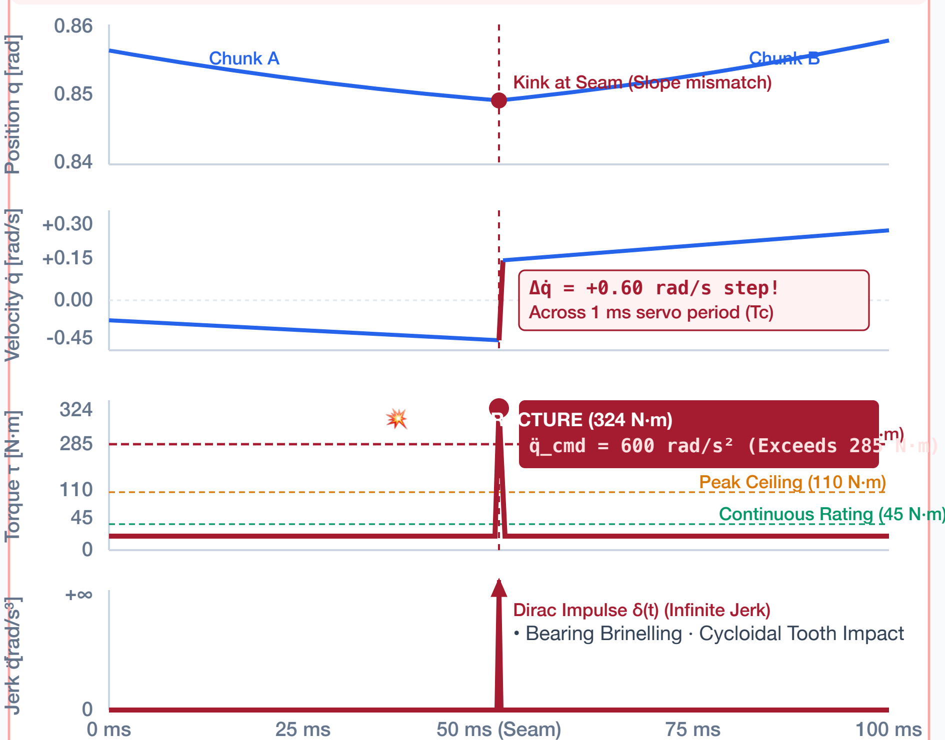
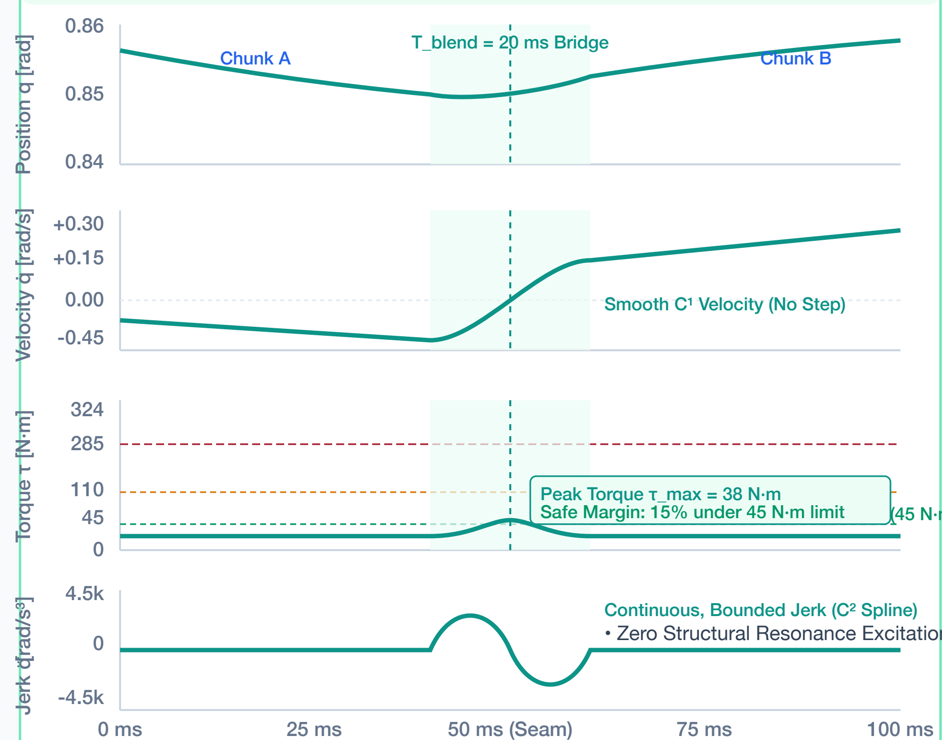
CASE A: UNCHECKED C^0 POSITION-ONLY CONCATENATION

C^0 FRACTURE



CASE B: JERK-CONTINUOUS C^2 QUINTIC BLENDED SPLINE

C^2 VERIFIED



SYSTEMS IMPLICATION: CONCATENATION REQUIRES HIGHER-ORDER BOUNDARY VALIDATION

When an action chunk finishes, splicing a replacement that matches only position (C^0) creates an instantaneous velocity step $\Delta\dot{q}$.

Across a 1.0 ms servo cycle (T_c), this step demands an acceleration impulse $\ddot{q} = \Delta\dot{q} / T_c$ = 600 rad/s², transmitting 324 N·m through reflected inertia (J = 2.7 kg·m²) and shearing the drive pin.